

| Question | Marking Guidance                                                                          | Mark           | Comments |
|----------|-------------------------------------------------------------------------------------------|----------------|----------|
| 06.1     | 3 cm <sup>3</sup> of (undiluted) disinfectant and 2 cm <sup>3</sup> of (distilled) water; | 1<br>(1 x AO2) |          |

| Question | Marking Guidance                                                                  | Mark           | Comments                                                                        |
|----------|-----------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------|
| 06.2     | 1. (Uncertainty =) 0.05 (cm <sup>3</sup> );<br>2. (Percentage uncertainty =) 2.5; | 2<br>(2 x AO2) | 2. Allow correct percentage uncertainty from student's incorrect answer to MP1. |

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mark               | Comments                                                                                                                                           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 06.3     | 1. Equally effective at 80% as at 100%;<br>2. Not effective at 20%<br><b>OR</b><br>Need higher than 20% concentration to be effective;<br>3. Light absorbance may not indicate number of living bacteria/cells<br><b>OR</b><br>Dead cells/bacteria absorb light<br><b>OR</b><br>Don't know if cells alive/dead at 80/100 %;<br>4. 100% not totally effective as still 10%/some absorbance;<br>5. Only (tested on) one species;<br><b>OR</b><br>Only tested on <i>B. subtilis</i> ;<br>6. Only tested in laboratory/test tube<br><b>OR</b><br>May not be as effective on other surfaces;<br>7. Only (tested) at 25°C; | 4 max<br>(4 x AO3) | 6. Accept any description which indicates not tested on non-living surfaces.<br><br>Ignore statistical test, sample size and reference to repeats. |